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Kite Pharma, Inc.
Our Ref.: 411-303 PCTEP

This is in reply to the Communication pursuant to Rules 161(1) and 162 EPC issued by the European Patent Office on September 13, 2023.

1. Requests

We submit an amended set of claims 1 to 12 and request that the European patent be granted on the basis thereof (clean and marked-up versions).

We shall adapt the description to the claims once the Examining Division has indicated that it intends to grant the European patent on the basis of the amended claims.

In case the Examining Division cannot accept the amended claims, we respectfully request the issuance of a Communication pursuant to Article 94(3) EPC or an informal telephone interview with the Representative. Should the Examining Division intend to refuse the application, oral proceedings in accordance with Article 116(1) EPC are requested.

2. Amendments; Article 123(2) EPC

Certain claims as originally filed have been combined in order to reduce the total number of claims. The following chart sets out the correspondence between the new claims submitted herewith (the amended set of claims) and the originally filed claims:

New claim	Originally filed claim(s)	New claim	Originally filed claim(s)
1	1, 3	7	17, 19
2	2, 4	8	18, 20
3	5 to 10	9	21, 22
4	11, 13	10	23, 24
5	12, 14	11	25, 26
6	15, 16	12	27

The subject-matter of the amended set of claims submitted herewith meets the requirements of Article 123(2) EPC.

3. Novelty; Article 54 EPC / Inventive Step; Article 56 EPC

The Written Opinion of the International Searching Authority alleges that the subject-matter of originally filed claims 1 to 27 is neither novel nor inventive. We respectfully disagree.

Claim 1, as amended herein, is directed to predicting the likelihood of survival of a patient in need of CAR T cell therapy based on the level of CD27+CD28+ naïve Th cells and the level of intermediate monocytes in an apheresis product obtained from the patient, wherein the patient has an increased likelihood of survival if the level of the CD27+CD28+ naïve Th cells in an apheresis product is over a cut-off percentage of total leukocytes and if the level of the intermediate monocytes is below a cut-off percentage of total leukocytes. Conversely, the patient has a decreased likelihood of survival if the level of the CD27+CD28+ naïve Th cells in an apheresis product is below a cut-off percentage of total leukocytes and if the level of the intermediate monocytes is above a cut-off percentage of total leukocytes.

This likelihood of survival is not based on the number of the CD27+CD28+ naïve Th cells or intermediate monocytes in the apheresis product but, rather, their percentage of all the leukocytes present in the apheresis product prior to processing of the product to produce T cells comprising a chimeric receptor. The likelihood of survival is also not just based the number of such cells immediately prior to administration or observed during treatment. Further, this likelihood of survival is for a patient in need of the CAR T cell therapy, not for a patient undergoing such therapy.

None of the cited prior art documents teach or suggest that the likelihood of survival of a patient in need of CAR T cell therapy can be predicted using a cut-off value based on the percentage of CD27+CD28+ naïve Th cells and intermediate monocytes in the apheresis product obtained from the patient. Nor is there any teaching or suggestion in the cited prior art documents regarding the other indicators for predicting a likelihood of survival of the patient, i.e., the level of CD27-CD28+ TEMRA Treg cells, the lymphocyte to leukocyte ratio, and/or the lymphocyte to monocyte ratio.

At best, the cited prior art teaches other cell populations that might be relevant for survival of the patient, such as in D6, or the number of infused CD8 and CCR7+CD45RA T cells after processing the apheresis product to produce T cells comprising a chimeric receptor, such as in D5. D4 discloses the expansion of certain population of cells and then administering the expanded population and that the number of expanded cells infused can be relevant for successful treatment. This is in in contrast to the presently claimed methods, which look at the percentage of CD27+CD28+ naïve Th cells and intermediate monocytes in the apheresis product obtained from the patient, absent any kind of expansion. D2 is directed to the absolute number of CD27+CD28+ naïve Th cells in the apheresis product obtained from the patient, not its percentage of the total number of leukocytes, not is there any suggestion of further taking into account the percentage of intermediate monocytes.

With regard to D6 being the closest prior art, we note that the difference is more than just a different cell population, D6 looks merely to the total number of cells belonging to the specified population, whereas in claim 1, the likelihood of survival is based on a cut-off value based on the percentage of cells of different populations to total leukocytes, not just CD27+CD28+ naïve Th cells. None of the cited prior art documents provide any pointer to population of CD27+CD28+ naïve Th cells and/or to intermediate monocytes, much less their percentage value to total leukocytes in the apheresis product obtained from the patient prior to processing.

We note that the foregoing inventive step analysis also applies to the other independent claims, new claims 4 and 7.

In view of the foregoing, we submit that the subject-matter of the amended set of claims submitted herewith is both novel and inventive within the meaning of Articles 54 and 56 EPC, respectively.

4. Conclusion

In summary, the amended set of claims submitted herewith meets all the requirements for patentability under the EPC.

Yours truly,



William Thomann
European Patent Attorney

Enclosure:
Amended set of claims 1 to 12 (clean and marked-up versions)